

SARA Advanced Physics Academy Organization

19 June 2008



Background

- Long extensive effort by RAAM has made a compelling case for the existence of the ARV/RAAM “Circuit”
 - RAAM is in possession of “Circuit” diagrams
 - The “Science” behind “Circuit” has not yet been demonstrated
- Progressive investors identified and ready to go
 - \$20-40M+ initial funding expected
- Bringing together all the players to get the project started has proven difficult
 - Issues include: IP, corporate structure, corporate control, development plan, business plan, ...
- This presentation proposes a framework based on the formation of two independent organizations



Definition: Technology Readiness Level (TRL)

Time Tested for Complex System Development

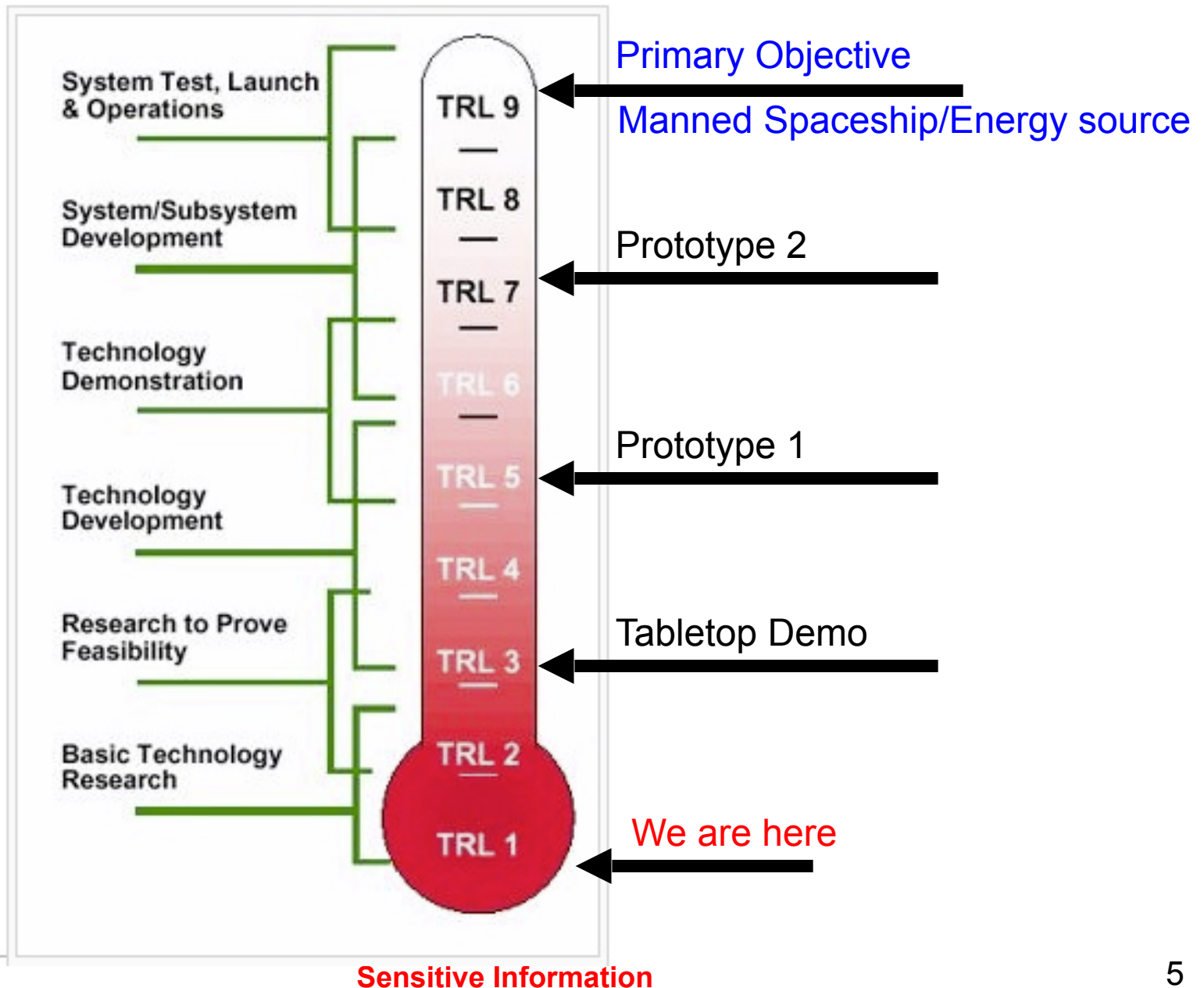
Technology Readiness Level	Description
1. Basic principles observed and reported	Lowest level of technology readiness. Scientific research begins to be translated into applied research and development. Example might include paper studies of a technology's basic properties.
2. Technology concept and/or application formulated	Invention begins. Once basic principles are observed, practical applications can be invented. The application is speculative and there is no proof or detailed analysis to support the assumption. Examples are still limited to paper studies.
3. Analytical and experimental critical function and/or characteristic proof of concept	Active research and development is initiated. This includes analytical studies and laboratory studies to physically validate analytical predictions of separate elements of the technology. Examples include components that are not yet integrated or representative.
4. Component and/or breadboard validation in laboratory environment	Basic technological components are integrated to establish that the pieces will work together. This is relatively "low fidelity" compared to the eventual system. Examples include integration of 'ad hoc' hardware in a laboratory.
5. Component and/or breadboard validation in relevant environment	Fidelity of breadboard technology increases significantly. The basic technological components are integrated with reasonably realistic supporting elements so that the technology can be tested in a simulated environment. Examples include 'high fidelity' laboratory integration of components.
6. System/subsystem model or prototype demonstration in a relevant environment	Representative model or prototype system, which is well beyond the breadboard tested for TRL 5, is tested in a relevant environment. Represents a major step up in a technology's demonstrated readiness. Examples include testing a prototype in a high fidelity laboratory environment or in simulated operational environment.
7. System prototype demonstration in an operational environment	Prototype near or at planned operational system. Represents a major step up from TRL 6, requiring the demonstration of an actual system prototype in an operational environment, such as in an aircraft, vehicle or space. Examples include testing the prototype in a test bed aircraft.
8. Actual system completed and 'flight qualified' through test and demonstration	Technology has been proven to work in its final form and under expected conditions. In almost all cases, this TRL represents the end of true system development. Examples include developmental test and evaluation of the system in its intended weapon system to determine if it meets design specifications.
9. Actual system 'flight proven' through successful mission operations	Actual application of the technology in its final form and under mission conditions, such as those encountered in operational test and evaluation. In almost all cases, this is the end of the last "bug fixing" aspects of true system development. Examples include using the system under operational mission conditions.

The Challenge Before Us

- TRL for the “Circuit/Rig” is currently in the 0-1 range!
 - *The underlying “Science” has not been identified or experimentally demonstrated*
- Thanks to the existence of the “Circuit” diagrams:
 - The development program can be accelerated once TRL 3 is achieved
- **Example: the Manhattan Project**
 - Fermi at University of Chicago demonstrated chain reaction (TRL 3)
 - The Manhattan Project was then launched to develop the A-Bomb based on Fermi’s demonstration (TRL 9 achieved in 3 years)
- **RAAM is missing “Fermi-like” compelling small-scale demonstration**
 - GHK’s “Science” is currently the leading candidate
 - Promises to achieve TRL 3+ in about 1 year
 - Other theories/hypothesis will also be considered given that compelling experimental demonstration can be shown within a year
 - It is vital for the identified “Science” to be advanced by a team (not a single individual) in order to accelerate development and reduce risk



Plan for Climbing TRL



Intermediate Specifications

TRL Level: 1 3 5 6 7 8 9

	Table Top	Proto 1 a,b,c,...	Proto 2 a,b,c,...	Full Scale a,b,c,...
Size	~ 1 ft ~ 1 lb	~ 3 ft ~ 10 lb	~ 10 ft ~ 100 lb	~ 50 ft ~ 2,500 lb
Power Source	External	External	On-board	On-board
Effect	~ 1% weight	~ 100% weight and ~ 1% power	> 100% weight and ~ 10% power	>> 100% weight and ~ 100% power
Control	On/Off	Basic Movement	Complex Movement	Remote/ Onboard
Test site	Laboratory	Parking lot	Remote site	White Sands, China Lake, ...



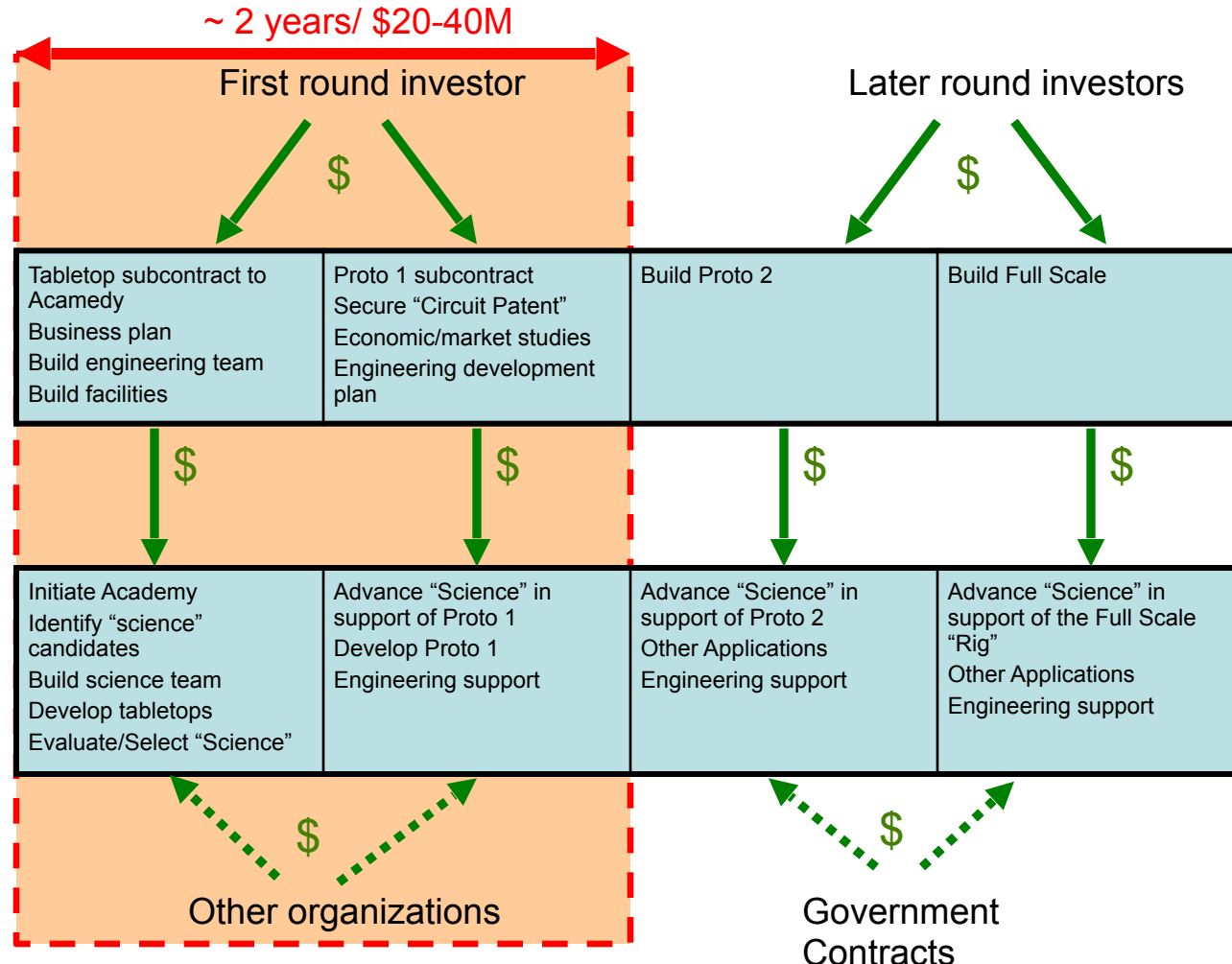
Suggested Approach:

Form Two Distinct Organizations

- **RAAM Organization**
 - Focused on the development and commercialization of the “Circuit”
 - Own “Circuit/Rig” IP
 - Responsible for evaluating and managing the development of the “Circuit/Rig”
 - Responsible for financing the project
- **Advanced Physics Academy Organization**
 - A Division of SARA
 - SARA can immediately provide access to services of 60+ highly trained scientists and engineers as well as lab facilities
 - Focused on the development of “Science” and the “Science” team
 - “Science” is not patentable, only its applications are patentable
 - Having a “team” is vital for accelerated development and lowered program risk
 - Primarily funded under subcontract to RAAM (or other organizations) in support of RAAM's “Circuit” development
 - Later activities may receive funding from government organizations
 - To pursue other applications of “Science”



Relationship Between the Two Organizations



RAAM

- "Circuit/Rig" IP, development & commercialization
- Financing
- Business plan management

SARA Advanced Physics Academy

- Develop "Science" and build the "Science" team
- Provide science and engineering services



Year 1-2 Activities (top level)

- **RAAM**
 - Send subcontract to the **Academy** for the development of “Tabletop” and “Prototype 1”
 - Secure “Circuit/Rig” Patent
 - Develop infrastructure for testing Prototype 2, and full scale “Rig”
 - Evolve detailed Business Plan to include engineering development and manufacturing
 - Perform economic impact study
- **SARA Advanced Physics Academy**
 - Establish a dedicated “Academy”, recruit key individuals and consultants
 - Expand theoretical, computer simulation, and Experimental capabilities
 - Perform a search for identifying and selecting qualifying “Science” candidates
 - Manage the development of Tabletop demo by the select “Science” candidates (TRL 2)
 - Manage the development of Prototype #1 using the preferred “Science” candidates who successfully demonstrated a Tabletop device (TRL 3+)
 - Provide science and engineering support to **RAAM**



Summary

- “Circuit” requires breakthrough “Science” not yet demonstrated
 - Promising “Science” candidates already identified
 - The breakthrough “Science” is essential for achieving TRL 3
 - SARA has the resources to start the “Science Academy” immediately
 - A highly focused engineering roadmap is possible for going from TRL 3 to TRL 9
- Formation of two organizations is recommended:
 - **RAAM** concentrates on “Circuit/Rig” development and commercialization
 - **SARA** advances science and provides low risk science and engineering support to **RAAM**
- Need key players gathered in a one-day meeting to work out the details and get the project started



SARA Advanced Physics Academy

Budget and Management Plan

19 June 2008



Academy Project and Budget Management Plan

- SARA has Program management system capable of managing large multi-million dollar projects
 - Microsoft Project based
 - Earned value system/Critical Path
 - Contract/Subcontract department
- SARA has government approved Job Cost system
 - Computerized task authorization and timekeeping
 - Government approved rates
- Expenditures for all tasks will be tracked and reported to RAAM and investors periodically
 - Tracked by task and subtasks number
 - All supporting data will be available for audit
- Invoices will be based on actual cost (Time & Material basis)
 - A retainer will be required at all times



Academy Work Breakdown Structure (WBS)

1. Program Management
 - Develop detailed plan and tracking
 - Periodical updates to the plan
 - Coordination with RAAM (quarterly technical and budget reviews)
 - Focus activities towards RAAM IP needs
 - Develop and enforce safety and security plans
2. Science Evaluation
 - Prepare RFP with specific requirements for the tabletop demo
 - Evaluate candidates using analysis and computer simulation
 - Select candidates for Tabletop development
 - GHK approach
 - JPD-led "Vortex" approach
 - Others
3. Prototype Development
 - Design and fabricate tabletop and prototypes
 - Evaluate using scientific methods
4. Expand Capability
 - People: hire scientists for Academy, or to backfill for current SARA technologist so they can support the Academy
 - Facilities: expand to new buildings as necessary, purchase lab dedicated instrumentation, computers, security needs, ...
5. Engineering Support
 - Provide analysis and fabrication services to RAAM engineering activities



Academy Expenditure Plan

- Objective: Tabletop and Prototype 1 demonstration in 2 years
 - Budget planned for maximum speed towards objective
 - Expenditure rate can be reduced but schedule will be stretched
- This plan will be updated quarterly based on progress

	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Total (\$K)
1. Program Management									
Develop detailed plan	120	40	40	40	40	40	40	40	400
Coordination with RAAM	15	15	15	15	15	15	15	15	120
Safety and security	50	50	50	50	50	50	50	50	400
Management Reserves	150	150	150	150	150	150	150	150	1,200
2. Science Evaluation									
Prepare & send RFPs	75	50							125
Evaluate candidates	50	150	150	100	100	50			600
3. Prototype Development									
Design & fabricate prototypes	300	450	450	450	600	600	600	600	4,050
Evaluate Prototypes		150	150	150	150	200	200	300	1,300
4. Expand Capability									
Recruit People:	150	150	150	150	150	150	150	150	1,200
Expand Facilities:	300	300	450	450	600	600	600	600	3,900
5. Engineering Support									
Provide services to RAAM		100	200	200	300	300	400	400	1,900
Total (\$K):	1,210	1,605	1,805	1,755	2,155	2,155	2,205	2,305	15,195



Science Evaluation Task Area

- Select “Candidates” based on latest intelligence
 - “Kirk” approach
 - “Vortex” approach (led by Dering)
 - ...
- Develop Request for Proposal (RFP) for candidates
 - Science must be described with references
 - How does it relate to RAAM “Circuit” geometry?
 - Can a Tabetop demo be designed and fabricated within a year?
 - 1% weight loss repeatable, reliable, and predictable
 - Must be able to disclose with Non-Disclosure Agreements
- Select and fund candidates
 - Within Academy
 - Subcontract to the candidate



Let's Get Started

- Time is of essence – why not get started now?
- Academy can get started immediately while RAAM structure and business plan are finalized
 - Academy will have no IP interest in RAAM
 - Can start with:
 - Develop RFP for “candidates”
 - GHK tabletop design
 - Recruit for talent and facility expansion options
- RAAM needs to get started ASAP to initiate economic studies and oversee Academy

